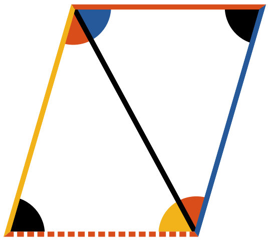

Book I. Prop. XXXIV. Theor. *The opposite sides and angles of any parallelogram are equal, and the diagonal ( ) divides it into two equal parts.*

$\because$ { = ; = } (pr. I.29)

and common to the two triangles.

$\therefore$ { = ; = ; = } (pr. I.26)

and = (ax. II).

Therefore the opposite sides and angles of the parallelogram are equal:

and as the triangles and are equal in every respect (pr. I.4), the diagonal divides the parallelogram into two equal parts.

Q. E. D.